

Hsien-Hsin Sean Lee, Ph.D.

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Intel Fellow | Editor-in-Chief, IEEE MICRO | IEEE Fellow



PROFESSIONAL PROFILE

Distinguished technology executive and computer architect with over 30 years of leadership across computer architecture, system software, semiconductors, and AI. Currently as an *Intel Fellow*, leads GPU and AI infrastructure pathfinding and product incubation for Intel's Data Center Group. Draws on extensive experience in both academia (Georgia Tech) and industry (Intel, TSMC, and Meta), with a proven track record of building and scaling world-class global R&D organizations.

PROFESSIONAL EXPERIENCE

Intel Corporation

Hudson, MA

Intel Fellow (Technology VP), AI System Architecture, Datacenter Group

Aug 2022 – Present

- Drive AI infrastructure technology pathfinding and product incubation.
- Lead CPU/GPU performance methodology and corporate strategy for AI workloads.
- Lead confidential AI and software stack strategy.
- Lead dynamic auto-tuning optimization frameworks.
- Key architect pushing architecture and emerging technology for datacenter GPU products.

Meta / Facebook, Inc.

Cambridge, MA

Area Research Lead, Facebook AI Research (FAIR)

March 2020 – Aug 2022

Research Head, AI Infrastructure Research

Jan 2019 – Feb 2020

- Established the "Systems for ML" research group from inception at the Facebook Boston site.
- Led R&D in high-efficiency personalized recommendation system designs.
- Led prototyping of processing-near-memory and emerging memory technology for AI workloads.
- Led research in sustainable, carbon-aware computing for machine learning.
- Led R&D for privacy-preserving machine learning.

Taiwan Semiconductor Manufacturing Company (TSMC)

Hsinchu, Taiwan

Deputy Director, Design Methodology and Kits Development Division

Apr 2012 – Jan 2019

- Directed 6 departments and 155 R&D staff across Taiwan, China, and the US developing IC design methodology, EDA tool features, and PDK for all TSMC customers for all nodes.
- Led IC design ecosystem enablement across TSMC, EDA vendors, and fabless design companies.
- Technology owner for all design collateral incl. advanced packaging (CoWoS, SoIC, WoW).
- Technology owner of machine learning applications for quality of physical verification.

Georgia Institute of Technology

Atlanta, GA

Associate Professor (Tenured), School of ECE

July 2008 – June 2014

Assistant Professor, School of ECE

July 2002 – June 2008

- Research program spanned multicore architecture, 3DIC, emerging memory technology, green datacenters, secure architecture, FPGA.
- Secured over \$5.8M (my own portion) in federal and industry funding.

Agere Systems

Atlanta, GA

Architecture Manager, StarCore DSP Technology Center

July 2001 – Aug 2002

- Managed 10 DSP architects from Agere and Motorola.
- Led StarCore DSP architecture development for 3G infrastructure.

Intel Corporation

Santa Clara / Folsom, CA

Senior Processor Architect / Microprocessor Research Labs Researcher

Oct 1995 – July 2001

- Pathfinding for next-gen IA64/EPIC architecture.
- Processor Architect of Pentium-III and Timna (early prototype of integrated CPU/GPU).
- Pentium III post-silicon validation.
- Coded and optimized Microsoft Direct3D 6.1 API using SSE ISA (assembly).

EDUCATION

Massachusetts Institute of Technology (MIT) <i>Master of Business Administration, Sloan School of Management</i>	Cambridge, MA 2026
University of Michigan <i>Ph.D. in Computer Science and Engineering</i> • Recipient of the Horace H. Rackham Distinguished Dissertation Award. • Nominated by University of Michigan for ACM Doctoral Thesis Award. <i>M.S.E. in Computer Science and Engineering</i>	Ann Arbor, MI 2001 1994
National Tsing-Hua University <i>B.S. in Electrical Engineering (Valedictorian of Class 1990)</i>	Hsinchu, Taiwan 1990

HONORS, SCHOLARSHIP, & SERVICES

- **IEEE Fellow:** Class of 2017, for contributions to 3D-ICs and Computer Architecture.
- **Distinguished Alumni Award:** EECS College, National Tsing-Hua University, 2026.
- **Hall of Fame:** Inducted into the MICRO and HPCA Halls of Fame.
- **Scholarly Impact:** Published 147 peer-reviewed papers; holder of 45 issued US Patents.
- **Awards:** 10-year Most Significant Paper (ITC 2017), NSF CAREER Award (2007), DoE CAREER Award (2005), and 3-time Top Picks papers in Computer Architecture (IEEE MICRO).
- **Editor-in-Chief:** IEEE MICRO (2024 – Present).
- **Program Chairs:** MICRO 2016, ISCA Industry Track 2021, SEED 2022.
- **General Chair:** IISWC 2010.

NOTABLE STUDENTS SUPERVISED AT GEORGIA TECH

- **Josh Fryman, Ph.D.** CTO & Fellow, Intel Government Technology, Intel Corp., OR.
- **Weidong (Larry) Shi, Ph.D.** Associate Professor, CS department, University of Houston, TX.
- **Taeweon Suh, Ph.D.** Professor, CSE department, Korea University, Seoul, South Korea.
- **Chinnakrishnan Ballapuram, Ph.D.** Sr. Director & HBF Base Die Chief Architect, Sandisk, CA.
- **Dong Hyuk Woo, Ph.D.** Sr. VP of AGI, Samsung, CA.
- **Mrinmoy Ghosh, Ph.D.** Sr. Director, HW/SW Co-Design, Samsung, CA.
- **Nak Hee Seong, Ph.D.** VP & Head of SOC IP Development, Samsung, South Korea.
- **Fayez Mohamood, M.S.** Founder and CEO of Bluecore (a unicorn startup), NY.
- **Richard Yoo, M.S.** CTO, Lunit, Seoul, South Korea.
- **Ilya (Khorosh) Tillis, B.S./M.S.** VP, Codility.
- **Greg Diamos, B.S.** Distinguished AI Engineer of multiple AI startups, CA.
- **June Paik (Joonho Baek), B.S.** Founder and CEO of Furiosa AI, South Korea/USA.

KEYNOTE SPEECH HIGHLIGHTS

- **Plenary Speaker.** "Technological Challenges of Social Network Computing" in International Conference on Electronics, Information, and Communication, Jeju, Korea, 2022.
- **Keynote Speaker.** "Challenges of Modern Computing on Social Network Platform" in Research Summit at the Center for Unstoppable Computing, University of Chicago, 2021.
- **NSF Keynote Speaker.** "When Memory Meets ML on Social Network Platform" in NSF Workshop on Processing-In-Memory Technology, National Science Foundation, 2021.
- **ECE Distinguished Lecture Series.** "Machine Learning on Social Network Platform" in George Washington University, Washington D.C., 2020.
- **CASPA Keynote Speaker.** "The Computing Frontiers of Social Network" in Chinese American Semiconductor Professional Association (CASPA) 2019 Annual Conference, Fremont, CA, 2019.
- **ISPLED Keynote Speaker.** "The Computing Frontiers of Social Network" in ACM/IEEE International Symposium on Low-Power Electronics and Design (ISLPED-2019), Lausanne Switzerland, 2019.
- **ISSCC Forum Speaker.** "Ecosystem of Design for Security" in Design Forum of the International Solid-State Circuits Conference (ISSCC-2016), CA, 2016.
- **Distinguished Speaker.** "3D Integration" in the Computer Information Science Engineering Department Colloquium, University of Florida, Gainesville, FL, 2010.

PATENTS

I hold **45 US issued patents** in the areas of machine learning, memory subsystem, secure non-volatile memory, 3D-IC, and physical design.

BOOK CHAPTERS

- BK1. Sungkap Yeo and Hsien-Hsin S. Lee. "[Peeling the Power Onion of Data Centers.](#)" Chapter Three In Energy Efficient Thermal Management of Data Centers by Yogendra Joshi and Pramod Kumar (Editors), pp.137-168, Springer, 2012.
- BK2. Abderrahim Benquassmi, Eric Fontaine, and Hsien-Hsin S. Lee. "[Parallelization of Katsevich CT Image Reconstruction Algorithm on Generic Multi-Core Processors and GPGPU.](#)" In GPU Computing GEMS, Section 10 Medical Imaging, Chapter 31, Wen-Mei Hwu (editor-in-chief), pp.659-677, Morgan Kaufmann Publishers, 2011.
- BK3. Intel 64 and IA-32 Architectures Optimization Reference Manual. (I was the original author of the Chapter of [Optimizing Cache Usage.](#))

JOURNAL & CONFERENCE PUBLICATION (BY AREAS)

Machine Learning Systems

- ML1. Wenjie Xiong, Liu Ke, Maxim Ostapenko, Yongmin Tai, Yeongon Cho, Joonho Song, Jinin So, Kyung-soo Kim, Yongsuk Kwon, Jin Jung, Jieun Lee, Byeongho Kim, Shin-haeng Kang, Sukhan Lee, Jeong-hyeon Cho, Kyomin Sohn, Xuan Zhang, Hsien-Hsin S. Lee, G. Edward Suh. "Accelerating Confidential Recommendation Model Inference with Near-Memory Processing." In IEEE Transactions on Dependable and Secure Computing, Vol 22, Issue 4, pg.3580-3586, 2025.
- ML2. Haiyang Huang, Newsha Ardalani, Anna Sun, Liu Ke, Shruti Bhosale, Hsien-Hsin S. Lee, Carole-Jean Wu, Benjamin C. Lee. "Toward Efficient Inference for Mixtures of Experts." In the *38th Annual Conference on Neural Information Processing System (NeurIPS'24)*, Vancouver, Canada, December 2024.
- ML3. Trishita Tiwari, Suchin Gururangan, Chuan Guo, Weizhe Hua, Sanjay Kariyappa, Udit Gupta, Wenjie Xiong, Kiwan Maeng, Hsien-Hsin S. Lee, G. Edward Suh. "Information Flow Control in Machine Learning Through Modular Model Architecture." In the *33rd USENIX Security Symposium*, Philadelphia, PA, August, 2024.
- ML4. Maximilian Lam, Jeff Johnson, Wenjie Xiong, Kiwan Maeng, Udit Gupta, Yang Li, Liangzhen Lai, Ilias Leontiadis, Minsoo Rhu, Hsien-Hsin S. Lee, Vijay Janapa Reddi, Gu-Yeon Wei, David Brooks, G. Edward Suh. "GPU-based Private Information Retrieval for On-Device Machine Learning." In the *ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS-2024)*, San Diego, California, April, 2024.
- ML5. Vivek Parmer, Syed Shakib Sarwar, Ziyun Li, Hsien-Hsin S. Lee, Barbara De Salvo, Manan Suri. "Exploring Memory-Oriented Design Optimization of Edge-AI Hardware for Extended Reality Applications." In *IEEE MICRO Special Issue on TinyML*, Volume 43, Issue 6, 2023.
- ML6. Sanjay Kariyappa, Chuan Guo, Kiwan Maeng, Wenjie Xiong, Edward Suh, Moinuddin K. Qureshi, and Hsien-Hsin S. Lee. "Cocktail Party Attack: Breaking Aggregation-based Privacy in Federated Learning Using Independent Component Analysis." In the *40th International Conference on Machine Learning (ICML'2023)*, Honolulu, Hawaii, July, 2023.
- ML7. Udit Gupta, Mariam Elgamal, Gage Hills, Gu-Yeon Wei, Hsien-Hsin S. Lee, David Brooks, Carole-Jean Wu. "Architectural CO₂ Footprint Tool: Designing Sustainable Computer Systems With an Architectural Carbon Modeling Tool." In *IEEE MICRO special issue on Top Picks from the Computer Architecture Conferences of 2022*, Volume 43, Issue 4, pp.107-117, 2023.
- ML8. Vivek Parmer, Syed Shakib Sarwar, Ziyun Li, Hsien-Hsin S. Lee, Barbara De Salvo, Manan Suri. "Memory-Oriented Design Space Exploration of Edge-AI Hardware for XR Applications." In *tinyML Research Symposium (tinyML-2023)*, 2023.

- ML9. Jiaxun Cui, Xiaomeng Yang, Mulong Luo, Geunbae Lee, Peter Stone, Hsien-Hsin S. Lee, Benjamin Lee, G. Edward Suh, Wenjie Xiong, Yuandong Tian. "MACTA: A Multi-agent Reinforcement Learning Approach for Cache Timing Attacks and Detection." In the *11th International Conference on Learning Representations (ICLR-2023)*, Kigali, Rwanda, May, 2023.
- ML10. Mulong Luo, Wenjie Xiong, Geunbae Lee, Yueying Li, Xiaomeng Yang, Amy Zhang, Yuandong Tian, Hsien-Hsin S. Lee, G. Edward Suh. "AutoCAT: Reinforcement Learning for Automated Exploration of Cache Timing-Channel Attacks." In *Proceedings of the 29th IEEE International Symposium on High-Performance Computer Architecture (HPCA-29)*, Montreal, QC, Canada, Feb, 2023.
- ML11. Hanieh Hashemi, Wenjie Xiong, Liu Ke, Kiwan Maeng, Murali Annavaram, G. Edward Suh, and Hsien-Hsin S. Lee. "Private Data Leakage via Exploiting Access Patterns of Sparse Features in Deep Learning-based Recommendation Systems." In the *2022 Trust and Socially Responsible Machine Learning* co-located with *NeurIPS*, December, 2022.
- ML12. Carole-Jean Wu, Ramya Raghavendra, Udit Gupta, Bilge Acun, Newsha Ardalani, Kiwan Maeng, Gloria Chang, Fiona Aga, James Huang, Charles Bai, Michael Gschwind, Anurag Gupta, Myle Ott, Anastasia Melnikov, David Brooks, Geeta Chauhan, Benjamin Lee, Hsien-Hsin S. Lee, Bugra Akyildiz, Maximilian Balandat, Joe Spisak, Ravi Jain, Mike Rabbat, Kim Hazelwood. "Sustainable AI: Environmental Implications, Challenges, and Opportunities." In the *Fifth Conference on Machine Learning and Systems (MLSys)*, August, 2022.
- ML13. Udit Gupta, Mariam Elgamal, Gage Hills, Gu-Yeon Wei, Hsien-Hsin S. Lee, David Brooks, and Carole-Jean Wu. "ACT: Designing Sustainable Computer Systems with an Architectural Carbon Footprint Modeling Tool." In *Proceedings of the 49th ACM/IEEE International Symposium on Computer Architecture (ISCA-49)*, pp.784-799, New York, New York, June, 2022.
- ML14. Yongqin Wang, Edward Suh, Wenjie Xiong, Benjamin Lefaudeaux, Brian Knot, Murali Annavaram, and Hsien-Hsin S. Lee. "Characterization of MPC-based Private Inferences for Transformer-based Models." In *2022 IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS-2022)*, pp.187-197, May, 2022.
- ML15. Udit Gupta, Young Geun Kim, Sylvia Lee, Jordan Tse, Hsien-Hsin S. Lee, Gu-Yeon Wei, David Brooks, Carole-Jean Wu. "Chasing Carbon: The Elusive Environmental Footprint of Computing." In *IEEE MICRO Top Picks from 2021 Computer Architecture Conferences*, pp.37-47, July-Aug, 2022.
- ML16. Wenjie Xiong, Liu Ke, Dimitrije Jankov, Michael Kounavis, Xiaochen Wang, Eric Northup, Jie Amy Yang, Bilge Acun, Carole-Jean Wu, Ping Tak Peter Tang, G. Edward Suh, Xuan Zhang, and Hsien-Hsin S. Lee. "SecNDP: Secure Near-Data Processing with Untrusted Memory." In the *28th IEEE International Symposium on High-Performance Computer Architecture (HPCA-28)*, pp. 244-258, Seoul, South Korea, April, 2022.
- ML17. Liu Ke, Udit Gupta, Mark Hempstead, Carole-Jean Wu, Hsien-Hsin S. Lee, and Xuan Zhang. "Hercules: Heterogeneity-aware Inference Serving for At-scale Personalized Recommendation." In the *28th IEEE International Symposium on High-Performance Computer Architecture (HPCA-28)*, pp. 141-154, Seoul, South Korea, April, 2022.
- ML18. Liu Ke, Xuan Zhang, Jinin So, Jong-Geon Lee, Shin-Haeng Kang, Sukhan Lee, Songyi Han, Yeongon Cho, Jin Hyun Kim, Yongsuk Kwon, Kyungsoo Kim, Jin Jung, Ilkwon Yun, Sung Joo Park, Hyunsun Park, Joonho Song, Jeonghyeon Cho, Kyomin Sohn, Nam Sung Kim, and Hsien-Hsin S. Lee. "Near-Memory Processing in Action: Accelerating Personalized Recommendation with AxDIMM." In *IEEE MICRO Special Issue on Processing-In-Memory*, January/February, 2022.
- ML19. Udit Gupta, Samuel Hsia, Jeff Zhang, Mark Wilkening, Javin Pombra, Hsien-Hsin S. Lee, Gu-Yeon Wei, Carole-Jean Wu, and David Brooks. "RecPipe: o-designing Models and Hardware to Jointly Optimize Recommendation Quality and Performance." In *Proceedings of the 54th ACM/IEEE International Symposium on Microarchitecture (MICRO-54)*, pp.870-884, Athens, Greece, October, 2021.
- ML20. Brandon Reagen, Woo-Seok Choi, Yeongil Ko, Vincent T. Lee, Hsien-Hsin S. Lee, Gu-Yeon Wei, David Brooks. "Cheetah: Optimizing and Accelerating Homomorphic Encryption for Private Inference." In *Proceedings of the IEEE International Symposium on High-Performance Computer Architecture (HPCA-27)*, Seoul, South Korea, February, 2021.

- ML21. Udit Gupta, Young Geun Kim, Sylvia Lee, Jordan Tse, Hsien-Hsin S. Lee, Gu-Yeon Wei, David Brooks, Carole-Jean Wu. "Chasing Carbon: The Elusive Environmental Footprint of Computing." In *Proceedings of the IEEE International Symposium on High-Performance Computer Architecture (HPCA-27)*, Seoul, South Korea, February, 2021.
- ML22. Udit Gupta, Samuel Hsia, Vikram Saraph, Xiaodong Wang, Brandon Reagen, Gu-Yeon Wei, Hsien-Hsin S. Lee, David Brooks, and Carole-Jean Wu. "DeepRecSys: A System for Optimizing End-to-End At-Scale Neural Recommendation Inference." In *Proceedings of the 47th ACM/IEEE International Symposium on Computer Architecture (ISCA-47)*, Valencia, Spain, 2020.
- ML23. Liu Ke, Udit Gupta, Carole-Jean Wu, Benjamin Cho, Mark Hempstead, Brandon Reagen, Xuan Zhang, David Brooks, Vikas Chandra, Utku Diril, Amin Firoozshahian, Bill Jia, Kim Hazelwood, Hsien-Hsin S. Lee, Meng Li, Bert Maher, Dheevatsa Mudigere, Maxim Naumov, Martin Scharz, Mikhail Smelyanskiy, and Xiaodong Wang. "RecNMP: Accelerating Personalized Recommendation with Near-Memory Processing." In *Proceedings of the 47th ACM/IEEE International Symposium on Computer Architecture (ISCA-47)*, Valencia, Spain, 2020.
- ML24. Udit Gupta, Carole-Jean Wu, Xiaodong Wang, Maxim Naumov, Brandon Reagen, David Brooks, Bradford Cottel, Kim Hazelwood, Mark Hempstead, Bill Jia, Hsien-Hsin S. Lee, Andrey Malevich, Dheevatsa Mudigere, Mikhail Smelyanskiy, Liang Xiong, and Xuan Zhang, "The Architectural Implications of Facebook's DNN-based Personalized Recommendation." In *Proceedings of the 26th IEEE International Symposium on High-Performance Computer Architecture (HPCA-26)*, San Diego, CA, February, 2020.
- ML25. Lifeng Nai, Yinglong Xia, Ching-Yung Lin, Bo Hong, and Hsien-Hsin S. Lee. "Cache-conscious Graph Collaborative Filtering on Multi-socket Multicore Systems". In *Proceedings of the 11th ACM Conference on Computing Frontiers*, Article No. 32, Cagliari, Italy, 2014.
- ML26. Richard M. Yoo, Han Lee, Kingsum Chow, and Hsien-Hsin S. Lee, "Constructing a Non-Linear Model with Neural Networks For Workload Characterization." In *Proceedings of the 2006 IEEE International Symposium on Workload Characterization (IISWC-06)*, pp.150-159, San Jose, California, October, 2006.

Datacenter, Cloud, and Multi-core Computing

- DC1. Sungkap Yeo, Mohammad M. Hossain, Jen-Cheng Huang, and Hsien-Hsin S. Lee. "ATAC: Ambient Temperature-Aware Capping for Power Efficient Datacenters." In *Proceedings of the ACM Symposium on Cloud Computing*, pp.17.1-17.14, Seattle, WA, 2014.
- DC2. Sungkap Yeo and Hsien-Hsin S. Lee. "Peeling the Power Onion of Data Centers." Chapter Three In *Energy Efficient Thermal Management of Data Centers* by Yogendra Joshi and Pramod Kumar (Editors), pp.137-168, Springer, 2012.
- DC3. Mohammad M. Hossain, Jen-Cheng Huang, and Hsien-Hsin S. Lee. "Migration Energy-Aware Workload Consolidation in Enterprise Clouds." In *Proceedings of the IEEE International Conference on Cloud Computing Technology and Science*, pp.405-410, December, 2012.
- DC4. Sungkap Yeo and Hsien-Hsin S. Lee. "SimWare: A Holistic Warehouse-scale Computer Simulator." In *IEEE Computer Special Issue on Modeling and Simulation of Smart and Green Computing Systems*, Vol. 35, No. 9, pp.48-55, 2012.
- DC5. Sungkap Yeo and Hsien-Hsin S. Lee. "Using Mathematical Modeling in Provisioning a Heterogeneous Cloud Computing Environment." In *IEEE Computer*, Vol. 44, No. 8, pp. 55-62, August, 2011.
- DC6. Mrinmoy Ghosh, Ripal Nathuji, Min Lee, Karsten Schwan, and Hsien-Hsin S. Lee. "Symbiotic Scheduling for Shared Caches in Multi-Core Systems Using Memory Footprint Signature." In *Proceedings of the 40th IEEE International Conference on Parallel Processing (ICPP-2011)*, Taipei, Taiwan, pp.11 - 20, September, 2011. (Acceptance rate = 22%, 81/363)
- DC7. Sung Woo Chung, Hsien-Hsin S. Lee, and Woo Hyong Lee. "Architecture/OS Support for Embedded Multi-core Systems." In *The Computer Journal*, vol.53, no.8, pp.1134-1135, 2010.

- DC8. Dong Hyuk Woo and Hsien-Hsin S. Lee. "PROPHET: Goal-Oriented Provisioning for Highly Tunable Multicore Processors in Cloud Computing." In *ACM SIGOPS Operating Systems Review special issue on the Interaction among the OS, Compilers, and Multicore Processors*, Vol.43, Issue 2, pp.102-103, April, 2009.
- DC9. Michael Healy, Hsien-Hsin S. Lee, Gabriel H. Loh, and Sung Kyu Lim. "Thermal optimization in Multigranularity Multi-Core Floorplanning." In *Proceedings of the 14th IEEE/ACM Asia South Pacific Design Automation Conference (ASP-DAC'09)*, pp.43-48, Yokohama, Japan, January, 2009.
- DC10. Dong Hyuk Woo and Hsien-Hsin S. Lee. "Extending Amdahl's Law for Energy-Efficient Computing in the Many-Core Era." In *IEEE Computer*, Vol.41, No.12, pp.24-31, December, 2008.
- DC11. Richard M. Yoo and Hsien-Hsin S. Lee, "Adaptive Transaction Scheduling for Transactional Memory Systems." In *Proceedings of the 20th ACM Symposium on Parallelism in Algorithms and Architectures (SPAA) in the Special Track on Hardware and Software Techniques to Improve the Programmability of Multicore Machines*, pp. 169-178, Munich, Germany, 2008.
- DC12. Richard M. Yoo, Yang Ni, Adam Welc, Bratin Saha, Ali-Reza Adl-Tabatabai, and Hsien-Hsin S. Lee, "Kicking the Tires of Software Transactional Memory: Why the Going Gets Tough." In *Proceedings of the 20th ACM Symposium on Parallelism in Algorithms and Architectures (SPAA) in the Special Track on Hardware and Software Techniques to Improve the Programmability of Multicore Machines*, pp.265-274, Munich, Germany, 2008.
- DC13. Richard M. Yoo and Hsien-Hsin S. Lee. "Helper Transactions: Enabling Thread-Level Speculation via A Transactional Memory System." In *Workshop on Parallel Execution of Sequential Programs on Multi-core Architectures (PESPMA) in Conjunction with ACM/IEEE International Symposium on Computer Architecture (ISCA-35)*, Beijing, China, June 2008.
- DC14. Taeweon Suh, Daehyun Kim, and Hsien-Hsin S. Lee, "Cache Coherence Support for Non-Shared Bus Architecture on Heterogeneous MP SoCs." In *Proceedings of the 42nd Design Automation Conference (DAC-42)*, pp.553-558, Anaheim, California, June 2005. (acceptance rate=20%)
- DC15. Taeweon Suh, Hsien-Hsin S. Lee, and Douglas M. Blough, "Integrating Cache Coherence Protocols for Heterogeneous Multiprocessor Systems. Part 2." In *IEEE MICRO special issue on Embedded Systems: Architecture, Design and Tools*, pp.70-78, September/October 2004.
- DC16. Taeweon Suh, Hsien-Hsin S. Lee, and Douglas M. Blough, "Integrating Cache Coherence Protocols for Heterogeneous Multiprocessor Systems. Part 1." In *IEEE MICRO special issue on Embedded Systems: Architecture, Design and Tools*, pp.33-41, July/August 2004.
- DC17. Taeweon Suh, Douglas M. Blough and Hsien-Hsin S. Lee, "Supporting Cache Coherence in Heterogeneous Multiprocessor Systems." In *Proceedings of the Design Automation and Test in Europe Conference (DATE'04)*, pp.1150-1155, Paris, France, February 2004. (acceptance rate = 23.2%, 181/780)

GPU and Heterogeneous Architecture

- GP1. Jen-Cheng Huang, Joo Hwan Lee, Hyesoon Kim, and Hsien-Hsin S. Lee. "GPUMech: GPU Performance Modeling Technique Based on Interval Analysis." In *Proceedings of the 47th IEEE/ACM International Symposium on Microarchitecture (MICRO-47)*, pp.268-279, Cambridge, England, December, 2014.
- GP2. Jen-Cheng Huang, Lifeng Nai, Hyesoon Kim, and Hsien-Hsin S. Lee. "TBPoint: Reducing Simulation Time for Large Scale GPGPU Kernels." In *Proceedings of the 28th International Parallel and Distributed Processing Symposium (IPDPS)*, pp.437-446, Phoenix, AZ, 2014.
- GP3. Hong Jun Choi, Dong Oh Son, Seung Gu Kang, Jong Myon Kim, Hsien-Hsin S. Lee, and Cheol Hong Kim. "An Efficient Scheduling Scheme Using Estimated Execution Time for Heterogeneous Computing Systems." In *Journal of Supercomputing*, 65(2), pp.886-902, 2013.
- GP4. Abderrahim Benquassmi, Eric Fontaine, and Hsien-Hsin S. Lee. "Parallelization of Katsevich CT Image Reconstruction Algorithm on Generic Multi-Core Processors and GPGPU." In *GPU Computing GEMS, Section 10 Medical Imaging, Chapter 31*, Wen-Mei Hwu (editor-in-chief), pp.659-677, Morgan Kaufmann Publishers, 2011.

- GP5. Dong Hyuk Woo and Hsien-Hsin S. Lee. "COMPASS: A Programmable Data Prefetcher Using Idle GPU Shaders." In *Proceedings of the 16th IEEE International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS XV)*, pp.297-309, Pittsburgh, PA, March, 2010. (Acceptance rate = 17.7%, 32/181)
- GP6. Ahmad Sharif and Hsien-Hsin S. Lee. "Total Recall: A Debugging Framework for GPUs." In *Proceedings of the ACM SIGGRAPH/Eurographics Workshop of Graphics Hardware (GH-08)*, pp.13-20, Sarajevo, Bosnia-Herzegovina, June, 2008.
- GP7. Eric Fontaine and Hsien-Hsin S. Lee, "Optimizing Katsevich Image Reconstruction Algorithm on Multicore Processors." In *Proceedings of the 13th IEEE International Conference on Parallel and Distributed Systems (ICPADS-07)*, Hsinchu, Taiwan, December, 2007.
- GP8. Dong Hyuk Woo, Joshua B. Fryman, Allan D. Knies, Marsha Eng, and Hsien-Hsin S. Lee, "POD: A Parallel On-Die Architecture." In *Proceedings of the 11th Annual Workshop on High Performance Embedded Computing (HPEC)*, Lexington, Massachusetts, September, 2007. (**Award Session**)
- GP9. Dong Hyuk Woo, Joshua B. Fryman, Allan D. Knies, and Hsien-Hsin S. Lee. "Chameleon: Virtualizing Idle Acceleration Cores of A Heterogeneous Multi-Core Processor for Caching and Prefetching." In *ACM Transactions on Architecture and Code Optimization*, vol.7, no.1, pp.1-35, April, 2010.

Energy Efficient Design

- EE1. Mrinmoy Ghosh, Simon Ford, Emre Ozer, Stuart Biles, and Hsien-Hsin S. Lee. "Way Guard: A Segmented Counting Bloom Filter Approach to Reducing Energy for Set-Associative Caches." In *Proceedings of the International Symposium on Low Power Electronics and Design (ISLPED-09)*, pp.165-170, San Francisco, CA, August, 2009. (**Selected as one of seven papers of ISLPED highlight for publicity and press.**)
- EE2. Hrishikesh Amur, Ripal Nathuji, Mrinmoy Ghosh, Karsten Schwan, and Hsien-Hsin S. Lee. "Idle-Power: Application-Aware Management of Processor Idle States." In *Workshop on Managed Many-Core Systems (MMCS) co-located with ACM/IEEE International Symposium on High Performance Distributed Computing (HPDC)*, Boston, MA, June, 2008.
- EE3. Chinnakrishnan S. Ballapuram, Ahmad Sharif, and Hsien-Hsin S. Lee, "Exploiting Access Semantics and Program Behavior to Reduce Snoop Power in Chip Multiprocessors." In *Proceedings of the 13th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS XIII)*, pp.60-69, Seattle, WA, 2008. (acceptance rate = 31/127, 24.4%)
- EE4. Chinnakrishnan S. Ballapuram and Hsien-Hsin S. Lee, "Improving TLB Energy for Java Applications on JVM." In *Proceedings of the International Symposium on Systems, Architectures, Modeling and Simulation (SAMOS VIII)*, pp.218-223, Samos, Greece, 2008.
- EE5. Mrinmoy Ghosh and Hsien-Hsin S. Lee, "Virtual Exclusion: An Architectural Approach to Reducing Leakage Energy in Caches for Multiprocessor Systems." In *Proceedings of the 13th IEEE International Conference on Parallel and Distributed Systems (ICPADS-07)*, Hsinchu, Taiwan, December, 2007.
- EE6. Dong Hyuk Woo, Mrinmoy Ghosh, Emre Ozer, Stuart Biles and Hsien-Hsin S. Lee, "Reducing Energy of Virtual Cache Synonym Lookup using Blooming Filters." In *Proceedings of the International Conference on Compilers, Architecture, Synthesis for Embedded Systems (CASES'06)*, pp.179-189, Seoul, Korea, October, 2006. (Regular paper acceptance rate = 24.3%, 25/103)
- EE7. Chinnakrishnan S. Ballapuram, Kiran Puttaswamy, Gabriel H. Loh and Hsien-Hsin S. Lee, "Entropy-based Low Power Data TLB Design." In *Proceedings of the International Conference on Compilers, Architecture, Synthesis for Embedded Systems (CASES'06)*, pp.304-311, Seoul, Korea, October, 2006. (Short paper acceptance rate = 39.8%, 41/103)
- EE8. Mrinmoy Ghosh and Hsien-Hsin S. Lee, "DRAMdecay: Using Decay Counters to Reduce Energy Consumption in DRAMs." In *Proceedings of the 3rd Watson Conference on Interaction between Architecture, Circuits and Compilers (PAC²)*, Yorktown Heights, NY, October, 2006.
- EE9. Mrinmoy Ghosh, Emre Ozer, Stuart Biles, and Hsien-Hsin S. Lee, "Efficient System-on-Chip Energy Management with a Segmented Bloom Filter." In *Proceedings of the Architecture of Computing Systems*

(ARCS'05), pp.283-297, Frankfurt, Germany, March 2006. (acceptance rate = 21%)

- EE10. Chinnakrishnan S. Ballapuram, Hsien-Hsin S. Lee, and Milos Prvulovic, "Synonymous Address Compaction for Energy Reduction in Data TLB." In *Proceedings of the ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED-05)*, pp.357-362, San Diego, California, August 2005. (acceptance rate=22%, 53/233)
- EE11. Mrinmoy Ghosh, Weidong Shi, and Hsien-Hsin S. Lee, "CoolPression — A Hybrid Significance Compression Technique for Reducing Energy in Caches." In *Proceedings of the IEEE International System-On-Chip Conference (SOCC-2004)*, pp. 399 - 402, Santa Clara, California, September, 2004.
- EE12. Yuvraj S. Dhillon, Abdulkadir U. Diril, Abhijit Chatterjee, and Hsien-Hsin S. Lee, "An Algorithm for Achieving Minimum Energy Consumption in CMOS Circuits Using Multiple Supply and Threshold Voltages at the Module Level." In *Digest of Technical Papers of the International Conference on Computer-Aided Design (ICCAD-03)*, pp.693-700, San Jose, California, November 2003. (acceptance rate = 26%, 130/490)
- EE13. Joshua B. Fryman, Chad M. Huneycutt, Hsien-Hsin S. Lee, Kenneth M. Mackenzie and David E. Schimmel, "Energy Efficient Network Memory for Ubiquitous Devices." In *IEEE MICRO special issue on Power Complexity Aware Design*, pp. 60-70, September/October 2003.
- EE14. Hsien-Hsin S. Lee and Chinnakrishnan S. Ballapuram, "Energy Efficient D-TLB and Data Cache using Semantic-Aware Multilateral Partitioning." In *Proceedings of the International Symposium on Low Power Electronics and Design (ISLPED'03)*, pp. 306-311, Seoul, Korea, August 2003. (acceptance rate = 24.4%, 54/221)
- EE15. Hsien-Hsin S. Lee, Joshua B. Fryman, A. Utku Diril, and Yuvraj S. Dhillon, "The Elusive Metric for Low-Power Architecture Research." In *Workshop on Complexity-Effective Design (WCED-03) held in conjunction with the 30th ACM/IEEE International Symposium on Computer Architecture (ISCA-30)*, San Diego, California, June 2003. (acceptance rate=25%)
- EE16. Hsien-Hsin S. Lee and Gary S. Tyson, "Region-based Caching: an Energy-Delay Efficient Memory Architecture for Embedded Processors." In *Proceedings of the International Conference on Compilers, Architecture, and Synthesis for Embedded Systems (CASES'00)*, pp.120-127, San Jose, California, November, 2000.

Memory Systems

- MM1. Nak Hee Seong, Sungkap Yeo, and Hsien-Hsin S. Lee. "Tri-Level-Cell Phase Change Memory: Toward an Efficient and Reliably Memory System." In *Proceedings of the 40th International Symposium on Computer Architecture (ISCA-40)*. Pp.440-451, Tel-Aviv, Israel, June, 2013.
- MM2. Sungkap Yeo, Nak Hee Seong, and Hsien-Hsin S. Lee. "Can Multi-Level Cell PCM Be Reliable and Usable? Analyzing the Impact of Resistance Drift." In the 10th Annual Workshop on Duplicating, Deconstructing and Debunking in conjunction with the 39th International Symposium on Computer Architecture, Portland, OR, June, 2012.
- MM3. Nak Hee Seong, Dong Hyuk Woo, Vijayalakshmi Srinivasan, Jude A. Rivers, and Hsien-Hsin S. Lee. "SAFER: Stuck-At-Fault Error Recovery for Memories." In *Proceedings of the 43rd ACM/IEEE International Symposium on Microarchitecture (MICRO-43)*, pp.115-124, Atlanta, Georgia, December, 2010. (Acceptance rate = 18%, 45/248)
- MM4. Ahmad Sharif and Hsien-Hsin S. Lee, "Data Prefetching by Exploiting Global and Local Access Patterns." In *Journal of Instruction-Level Parallelism, Special Issue: The First JILP Data Prefetching Championship (DPC-1)*, Volume 13, 2011. ISSN 1942-9525.
- MM5. Nak Hee Seong, Dong Hyuk Woo, and Hsien-Hsin S. Lee. "Security Refresh: Prevent Malicious Wear-out and Increase Durability for Phase-Change Memory with Dynamically Randomized Address Mapping." In *IEEE MICRO special issue on Top Picks from the Computer Architecture Conferences of 2010*, pp.119-127, January/February, 2011.
- MM6. Nak Hee Seong, Dong Hyuk Woo, and Hsien-Hsin S. Lee. "Security Refresh: Prevent Malicious Wear-out and Increase Durability for Phase-Change Memory with Dynamically Randomized Address

Mapping." In *Proceedings of the 37th IEEE International Symposium on Computer Architecture (ISCA-37)*, pp.383-394, Saint-Malo, France, June, 2010. (Acceptance rate = 18%, 44/245)

- MM7. Ahmad Sharif and Hsien-Hsin S. Lee. "Data Prefetching Mechanism by Exploiting Global and Local Access Patterns" In *The First Journal of Instruction-Level Parallelism Data Prefetching Championship (DPC-1) in conjunction with the 15th IEEE International Symposium on High Performance Computer Architecture (HPCA-15)*, Raleigh, North Carolina, February, 2009.
- MM8. Mrinmoy Ghosh and Hsien-Hsin S. Lee, "Smart Refresh: An Enhanced Memory Controller Design for Reducing Energy in Conventional and 3D Die-Stacked DRAMs." In *Proceedings of the 40th IEEE/ACM International Symposium on Microarchitecture (MICRO-40)*, pp.134-145, Chicago, IL, December, 2007. (acceptance rate = 35/166 = 21%)
- MM9. Xiaotong Zhuang and Hsien-Hsin S. Lee, "Reducing Cache Pollution via Dynamic Data Prefetch Filtering." In *IEEE Transactions on Computers*, Vol.56, No.1, pp.18-31, January, 2007.
- MM10. Xiaotong Zhuang and Hsien-Hsin S. Lee, "A Hardware Based Cache Pollution Filtering Mechanism for Aggressive Prefetches." In *Proceedings of the International Symposium on Parallel Processing (ICPP-03)*, pp.286-293, Kaohsiung, Taiwan, October 2003. (acceptance rate = 20% in architecture track)
- MM11. Hsien-Hsin S. Lee, Gary S. Tyson, and Matthew K. Farrens, "Bandwidth Utilization using Eager Write-back." *Journal of Instruction-Level Parallelism*, Vol. 4, 2001.
- MM12. Hsien-Hsin S. Lee, Gary S. Tyson, and Matthew K. Farrens, "Eager Writeback - a Technique for Improving Bandwidth Utilization." In *Proceedings of the 33rd ACM/IEEE International Symposium on Microarchitecture (MICRO-33)*, pp.11-21, Monterey, California, December, 2000. (**Best Paper Award of MICRO-33.**)
- MM13. *Intel Architecture Software Optimization Reference Manual*, Intel Literature Center, order number: 245127-001, August, 1998. (Author of Chapter 6: Optimizing Cache Utilization for Pentium III Processor, pp.6-1 to pp.6-30, and Appendix A: The Mathematics of Prefetch Scheduling Distance, pp.F-1 to pp.F-12.)

3DIC Design and Test

- 3D1. Tianjian Li, Yan Han, Hsien-Hsin S. Lee, and Li Jiang. "Fault Clustering Technique for 3D Memory BISR," In *ACM/IEEE Design Automation & Test in Europe*, Lausanne, Switzerland, March, 2017.
- 3D2. Daehyun Kim, Krit Athikulwongse, Michael B. Healy, Mohammad M. Hossain, Moongon Jung, Ilya Khorosh, Gokul Kumar, Young-Joon Lee, Dean L. Lewis, Tzu-Wei Lin, Chang Liu, Shreepad Panth, Mohit Pathak, Minzhen Ren, Guanhao Shen, Taigon Song, Dong Hyuk Woo, Xin Zhao, Joungho Kim, Ho Choi, Gabriel H. Loh, Hsien-Hsin S. Lee, Sung Kyu Lim. "Design and Analysis of 3D-MAPS (3D Massively Parallel Processor with Stacked Memory." In *IEEE Transactions on Computers*, 64(1), pp.112-125, 2015.
- 3D3. Dong Hyuk Woo, Nak Hee Seong, and Hsien-Hsin S. Lee. "Pragmatic Integration of An SRAM Row Cache in Heterogeneous 3D DRAM Architecture using TSV." In *IEEE Transactions on Very Large Scale Integrated Circuits and Systems*, Vol.21, No.1, pp.1-13, January 2013.
- 3D4. Dae Hyun Kim, Krit Athikulwongse, Michael B. Healy, Mohammad M. Hossain, Moongon Jung, Ilya Khorosh, Gokul Kumar, Young-Joon Lee, Dean L. Lewis, Tzu-Wei Lin, Chang Liu, Shreepad Panth, Mohit Pathak, Minzhen Ren, Guanhao Shen, Taigon Song, Dong Hyuk Woo, Xin Zhao, Joungho Kim, Ho Choi, Gabriel H. Loh, Hsien-Hsin S. Lee, and Sung Kyu Lim. "3D-MAPS: 3D Massively Parallel Processor with Stacked Memory." In *Technical Digest of the IEEE International Solid-State Circuits Conference (ISSCC)*, pp.188 - 190, San Francisco, CA, 2012.
- 3D5. Hong Jun Choi, Young Jin Park, Hsien-Hsin Lee, and Cheol Hong Kim. "Adaptive Dynamic Frequency Scaling for Thermal-Aware 3D Multi-core Processors." In the *Proceedings of the 12th International Conference on Computational Science and Its Applications*, pp.602-612, Salvador de Bahia, Brazil, 2012.
- 3D6. Xiaodong Wang, Dilip Vasudevan, and Hsien-Hsin S. Lee. "Global Built-In Self-Repair for 3-D Memories with Redundancy Sharing and Parallel Testing." In *IEEE International 3D System Integration*

Conference (3DIC-12), Osaka, Japan, 2012.

- 3D7. Dean L. Lewis, Shreepad Panth, Xin Zhao, Sung Kyu Lim, and Hsien-Hsin S. Lee. "Designing 3D Test Wrappers for Pre-bond and Post-bond Test of 3D Embedded Cores." In *Proceedings of the XXIX IEEE International Conference on Computer Design (ICCD-11)*, pp.90 - 95, University of Massachusetts, Amherst, USA, October, 2011.
- 3D8. Xin Zhao, Dean Lewis, Hsien-Hsin S. Lee, and Sung Kyu Lim, "Low-Power Clock Tree Design for Pre-Bond Testing of 3D Stacked ICs." In *IEEE Transactions on Computer Aided Design of Integrated Circuits and Systems*, Vol.30, No.5, pp.732-745, 2011.
- 3D9. Dong Hyuk Woo, Nak Hee Seong, and Hsien-Hsin S. Lee. "Heterogeneous Die Stacking of SRAM Row Cache and 3-D DRAM: An Empirical Design Evaluation." In *Proceedings of the 54th IEEE International Midwest Symposium on Circuits and Systems*, pp.1 - 4, Seoul, Korea, August, 2011. (An invited paper.)
- 3D10. Dong Hyuk Woo, Nak Hee Seong, Dean L. Lewis, and Hsien-Hsin S. Lee. "An Optimized 3D-Stacked Memory Architecture by Exploiting Excessive, High-Density TSV Bandwidth." In *Proceedings of the 16th IEEE International Symposium on High-Performance Computer Architecture (HPCA-16)*, pp.429-440, Bangalore, India, January, 2010. (Acceptance rate = 18%)
- 3D11. Michael B. Healy, Krit Athikulwongse, Rohan Goel, Mohammad M. Hossain, Dae Hyun Kim, Young-Joon Lee, Dean L. Lewis, Tzu-Wei Lin, Chang Liu, Moongon Jung, Brian Ouellette, Mohit Pathak, Hemant Sane, Guan hao Shen, Dong Hyuk Woo, Xin Zhao, Gabriel H. Loh, Hsien-Hsin S. Lee, and Sung Kyu Lim. "Design and Analysis of 3D-MAPS: A Many-Core 3D Processor with Stacked Memory." In *Proceedings of the IEEE Custom Integrated Circuits Conference (CICC)*, San Jose, California, September, 2010.
- 3D12. Dean Lewis, Michael Healy, Mohammad Hossain, Tzu-Wei Lin, Mohit Pathak, Hemant Sane, Sung Kyu Lim, Gabriel Loh, and Hsien-Hsin S. Lee. "Design and test of 3D-MAPS, a 3D Die-Stack Many-Core Processor." In *the First IEEE International Workshop on Testing Three-Dimensional Stacked Integrated Circuits (poster)*, Austin, Texas, November, 2010.
- 3D13. Hsien-Hsin S. Lee and Krishnendu Chakrabarty. "Test Strategies for 3D Integrated Circuits." In *IEEE Design & Test of Computers, Special Issue on 3D IC Design and Test*, vol.26, no.5, pp.26-35, September/October, 2009.
- 3D14. Xin Zhao, Dean L. Lewis, Hsien-Hsin S. Lee, and Sung Kyu Lim. "Pre-bond Testable Low-Power Clock Tree Design for 3D Stacked ICs" In *Proceedings of the 2009 International Conference on Computer-Aided Design (ICCAD-09)*, pp.184-190, San Francisco, CA, November, 2009. (**Nominated for the Best Paper Award.**)
- 3D15. Dean L. Lewis and Hsien-Hsin S. Lee. "Architectural Evaluation of 3D Stacked RRAM Caches" In *Proceedings of the IEEE International 3D Systems Integration Conference (3DIC-09)*, pp.1-4. San Francisco, CA, September, 2009.
- 3D16. Dean L. Lewis and Hsien-Hsin S. Lee. "Testing Circuit-Partitioned 3D IC Designs." In *Proceedings of the IEEE Computer Society Annual Symposium on VLSI (ISVLSI-09)*, pp.139-144, Tampa, FL, May, 2009.
- 3D17. Dean L. Lewis, Sudhakar Yalamanchili, and Hsien-Hsin S. Lee. "High Performance Non-blocking Switch Design in 3D Die-Stacking Technology." In *Proceedings of the IEEE Computer Society Annual Symposium on VLSI (ISVLSI-09)*, pp.25-30, Tampa, FL, May, 2009.
- 3D18. Dong Hyuk Woo, Joshua B. Fryman, Allan D. Knies, Marsha Eng, and Hsien-Hsin S. Lee. "POD: A 3D-integrated Broad-Purpose Acceleration Layer." In *IEEE MICRO special issue on Accelerator Architectures*, pp.28-40, July/August, 2008.
- 3D19. Dean L. Lewis, and Hsien-Hsin S. Lee, "A Scan-Island Based Design Enabling Pre-bond Testability in Die-Stacking Microprocessors." In *Proceedings of the International Test Conference (ITC 2007)*, pp. 1-8, Santa Clara, CA, October, 2007. (**ITC-2017 10-year Most Significant Paper Award**)
- 3D20. Michael Healy, Mario Vittes, Mongkol Ekpanyapong, Chinnakrishnan Ballapuram, Sung Kyu Lim, Hsien-Hsin S. Lee, and Gabriel H. Loh, "Multi-Objective Microarchitectural Floorplanning For 2D and

3D ICs." In *IEEE Transactions on Computer Aided Design of Integrated Circuits and Systems*, Vol.26, No.1, pp.38-52, 2007.

Generic Microarchitecture

- uA1. Fayez Mohamood, Mrinmoy Ghosh, and Hsien-Hsin S. Lee. "DLL-Conscious Instruction Fetch Optimization for SMT Processors." In *Journal of Systems Architecture*, 54, pp.1089-1100, 2008.
- uA2. Eric Fontaine and Hsien-Hsin S. Lee. "Bicephaly: Maximizing Bandwidth by Duplexing Power and Data. In *Workshop on Wild and Crazy Ideas (WACI-VI) in conjunction with the International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS XIII)*, Seattle, WA, 2008.
- uA3. Fayez Mohamood, Michael Healy, Sung Kyu Lim, and Hsien-Hsin S. Lee, "A Floorplan-Aware Dynamic Inductive Noise Controller for Reliable Processor Design." In *Proceedings of the ACM/IEEE International Symposium on Microarchitecture (MICRO-39)*, pp. 3-14, Orlando, Florida, December, 2006. (acceptance rate = 24.1%, 42/174)
- uA4. Mongkol Ekpanyapong, Jacob Minz, Thaisiri Watwai, Hsien-Hsin S. Lee, and Sung Kyu Lim, "Profile-Guided Microarchitectural Floorplanning for Deep Submicron Processor Design." In *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, Vol.25, No.7, pp.1289-1300, July, 2006.
- uA5. Fayez Mohamood, Mrinmoy Ghosh and Hsien-Hsin S. Lee, "DLL-Conscious Instruction Fetch Optimization for SMT Processors." In *Proceedings of the 2nd Watson Conference on Interaction Between Architecture, Circuits and Compilers ($P = AC^2$)*, pp.143-152, Yorktown Heights, New York, September 2005. . **(Best Paper Award of $P = AC^2$.)**
- uA6. Mongkol Ekpanyapong, Sung Kyu Lim, Chinnakrishnan Ballapuram, and Hsien-Hsin S. Lee, "Wire-driven Microarchitectural Design Space Exploration," In *Proceedings of the 2005 IEEE International Symposium on Circuits and Systems (ISCAS-05)*, pp.1867-1870, Kobe, Japan, May 2005.
- uA7. Mongkol Ekpanyapong, Pinar Korkmaz, and Hsien-Hsin S. Lee, "Choice Predictor for Free." In *Proceedings of the 9th IEEE Asia-Pacific Computer Systems Architecture Conference (ACSAC-2004)*, pp.399 - 413, Beijing, China, September 2004.
- uA8. Mikhail Smelyanskiy, Scott A. Mahlke, Edward S. Davidson, and Hsien-Hsin S. Lee, "Predicate-aware Scheduling: A Technique for Reducing Resource Constraints." In *Proceedings of the First Annual IEEE/ACM International Symposium on Code Generation and Optimization (CGO-2003)*, pp.168-177, Fisherman's Wharf, San Francisco, California, March 2003. (acceptance rate = 35.4%)
- uA9. Hsien-Hsin S. Lee, Mikhail Smelyanskiy, Chris J. Newburn, and Gary S. Tyson, "Stack Value File: Custom Microarchitecture for the Stack." In *Proceedings of the 7th IEEE International Symposium on High Performance Computer Architecture (HPCA-7)*, pp.5-14, Monterrey, Mexico, January, 2001. (acceptance rate = 23.6%, 26/110)
- uA10. Paul Zagacki, Deep Buch, Emile Hsieh, Daniel Melaku, Vladimir Pentkovski, and Hsien-Hsin Lee, "Architecture of a 3D Software Stack for Peak Pentium III Processor Performance." *Intel Technology Journal*, Volume 3, Issue 2, May, 1999.

Physical Design

- PD1. Hsien-Hsin S. Lee, "IC Design Challenges and Opportunities for Advanced Process Technology," In *Proceedings of 2015 International Symposium on VLSI Design, Automation, and Test (VLSI-DAT)*, Hsinchu, Taiwan, April, 2015.
- PD2. Michael B. Healy, Fayez Mohamood, Hsien-Hsin S. Lee, and Sung Kyu Lim. "Integrated Microarchitectural Floorplanning and Runtime Controller for Inductive Noise Mitigation." In *ACM Transactions on Design Automation of Electronic Systems*, Volume 16, Issue 4, pp.46:1-25, 2011.
- PD3. Michael Healy, Fayez Mohamood, Hsien-Hsin S. Lee and Sung Kyu Lim, "A Unified Methodology for Power Supply Noise Reduction in Modern Microarchitecture Design." In *Proceedings of the 13th IEEE/ACM Asia South Pacific Design Automation Conference (ASP-DAC'08)*, pp.611-616, Seoul, Korea, 2008.

- PD4. Fayez Mohamood, Michael Healy, Sung Kyu Lim, and Hsien-Hsin S. Lee, "Noise-Direct: A Technique for Power Supply Noise Aware Floorplanning Using Microarchitecture Profiling." In *Proceedings of the 12th Asia and South Pacific Design Automation Conference (ASP-DAC'07)*, pp.786-791, Yokohama, Japan, January, 2007. (acceptance rate = 32.1%, 131/408)
- PD5. Michael Healy, Mario Vittes, Mongkol Ekpanyapong, Chinnakrishnan Ballapuram, Sung Kyu Lim, Hsien-Hsin S. Lee, and Gabriel H. Loh, "Microarchitectural Floorplanning Using Performance and Temperature Tradeoff." In *Proceedings of the Design, Automation and Test in Europe (DATE-06)*, pp.1288-1293, Munich, Germany, March 2006. (acceptance rate = 17%)
- PD6. Mongkol Ekpanyapong, Jacob R. Minz, Thaisiri Watwai, Hsien-Hsin S. Lee, and Sung Kyu Lim, "Profile-Guided Microarchitectural Floorplanning for Deep Submicron Processor Design." In *Proceedings of the 41st Design Automation Conference (DAC-2004)*, pp.634 -639, San Diego, California, June 2004. (acceptance rate = 21%, 163/785)

Security Architecture and Dependable Systems

- SA1. Jen-Cheng Huang, Matteo Monchiero, Yoshio Turner, and Hsien-Hsin S. Lee. "Ally: OS-Transparent Packet Inspection Using Sequestered Cores." In *Proceedings of the ACM/IEEE Symposium on Architectures for Networking and Communications Systems (ANCS-11)*, pp. 1 - 11, Brooklyn, NY, October, 2011. (Acceptance rate = 32%, 20/62) (**Best Paper Award of ANCS-11.**)
- SA2. Jun Yang, Lan Gao, Youtao Zhang, Marek Chrobak, and Hsien-Hsin S. Lee. "A Low-Cost Memory Remapping Scheme for Address Bus Protection" In *Journal of Parallel and Distributed Computing*, vol.70, issue 5, pp.443-457, May 2010.
- SA3. Vikas R. Vasisht and Hsien-Hsin S. Lee. "SHARK: Architectural Support for Autonomic Protection Against Stealth by Rootkit Exploits." In *Proceedings of the 41st ACM/IEEE International Symposium on Microarchitecture (MICRO-41)*, pp.106-116, Lake Como, Italy, November, 2008. (Acceptance rate = 19%, 40/210.)
- SA4. Weidong Shi and Hsien-Hsin S. Lee, "Accelerating Memory Decryption and Authentication with Frequent Value Prediction." In *Proceedings of the ACM International Conference on Computing Frontiers (CF'07)*, pp.35-46, Ischia, Italy, May, 2007.
- SA5. Weidong Shi, Chenghuai Lu, and Hsien-Hsin S. Lee, "Memory-centric Security Architecture." Invited article. In *Transactions of High Performance Embedded Architectures and Compilers*, Vol.1, pp.95-115, 2007. Springer Verlag.
- SA6. Dong Hyuk Woo and Hsien-Hsin S. Lee, "Analyzing Performance Vulnerability due to Resource Denial-of-Service Attack on Chip Multiprocessors." In *Workshop on Chip Multiprocessor Memory Systems and Interconnects (CMPMSI) in conjunction with the 13th International Symposium on High-Performance Computer Architecture (HPCA-13)*, Phoenix, Arizona, February, 2007.
- SA7. Weidong Shi, and Hsien-Hsin S. Lee, "Authentication Control Point and its Implications for Secure Processor Design." In *Proceedings of the ACM/IEEE International Symposium on Microarchitecture (MICRO-39)*, pp.103-112, Orlando, Florida, December, 2006. (acceptance rate = 24.1%, 42/174)
- SA8. Chenghuai Lu, Tao Zhang, Weidong Shi, and Hsien-Hsin S. Lee, "M-TREE: A High Efficiency Security Architecture for Protecting Integrity and Privacy of Software." In *Journal of Parallel and Distributed Computing, Special Issue on Security in Grid and Distributed Systems*, Vol.66, Issue 9, pp.1116-1128, 2006. (acceptance rate = 12.2%, 10/82.)
- SA9. Weidong Shi, Hsien-Hsin S. Lee, Richard M. Yoo, and Alexandra Boldyreva, "A Digital Rights Enabled Graphics Processing System." In *Proceedings of the ACM SIGGRAPH/Eurographics Workshop of Graphics Hardware*, pp.17-26, Vienna, Austria, September 2006. (acceptance rate = 31.1%, 14/45)
- SA10. Lan Gao, Jun Yang, Marek Chrobak, Youtao Zhang, San Nguyen, and Hsien-Hsin S. Lee, "A Low-cost Memory Remapping Scheme for Address Bus Protection." In *Proceedings of the 15th International Conference on Parallel Architecture and Compilation Techniques (PACT'06)*, pp.74-83, Seattle, WA, September 2006. (acceptance rate = 25.6%, 30/117)

- SA11. Weidong Shi, Hsien-Hsin S. Lee, Laura Falk, and Mrinmoy Ghosh, "An Integrated Framework for Dependable and Revivable Architecture Using Multicore Processors." In *Proceedings of the 33rd International Symposium on Computer Architecture (ISCA-33)*, pp.102-113, Boston, MA, June 2006. (acceptance rate = 13%, 31/231)
- SA12. Weidong Shi, Joshua B. Fryman, Guofei Gu, Hsien-Hsin S. Lee, Youtao Zhang, and Jun Yang, "InfoShield: A Security Architecture for Protecting Information Usage in Memory." In *Proceedings of the 12th International Conference on High Performance Computer Architectures (HPCA-12)*, pp.225-234, Austin, Texas, February 2006. (acceptance rate=14%)
- SA13. Weidong Shi, Hsien-Hsin S. Lee, Chenghuai Lu, and Mrinmoy Ghosh, "Towards the Issues in Architectural Support for Protection of Software Execution." In *ACM SIGARCH Computer Architecture News*, Vol. 33, Issue 1, pp.6-15, March 2005.
- SA14. Weidong Shi, Chenghuai Lu, and Hsien-Hsin S. Lee, "Memory-centric Security Architecture." In *Proceedings of the 2005 International Conference on High Performance Embedded Architectures and Compilers (HiPEAC 2005)*, pp.153 - 168, Barcelona, Spain, November 2005. (acceptance rate=20.2%, 17/84)
- SA15. Weidong Shi, Hsien-Hsin S. Lee, Guofei Gu, Mrinmoy Ghosh, Laura Falk, and Trevor Mudge, "Intrusion Tolerant and Self-recoverable Network Service System Using Security Enhanced Chip Multiprocessor." In *Proceedings of the 2nd IEEE International Conference on Autonomic Computing (ICAC-05)*, pp.263-273, Seattle, WA, June 2005. (acceptance rate of regular papers = 16.7%, 25/150)
- SA16. Weidong Shi, Hsien-Hsin S. Lee, Mrinmoy Ghosh, Chenghuai Lu, and Alexandra Boldyreva, "High Efficiency Counter Mode Security Architecture via Prediction and Precomputation." In *Proceedings of the 32nd International Symposium on Computer Architecture (ISCA-32)*, pp.14-24, Madison, Wisconsin, June 2005. (acceptance rate = 23.2%, 45/194)
- SA17. Weidong Shi, Hsien-Hsin S. Lee, Chenghuai Lu and Tao Zhang, "Attacks and Risk Analysis for Hardware Supported Software Copy Protection Systems." In *Proceedings of the 4th ACM Workshop on Digital Right Management (DRM'2004)*, pp.54-62, Washington D.C., October 2004.
- SA18. Weidong Shi, Hsien-Hsin S. Lee, Mrinmoy Ghosh, and Chenghuai Lu, "Architectural Support for High Speed Authentication of Shared Memory in Multiprocessor Systems." In *Proceedings of the International Conference on Parallel Architecture and Compilation Techniques (PACT'04)*, pp. 123 - 134, Antibes Juan-les-Pins, France, September 2004. (acceptance rate = 18.8%, 23/122)
- SA19. Xiaotong Zhuang, Tao Zhang, Hsien-Hsin S. Lee and Santosh Pande, "Hardware Assisted Control Flow Obfuscation for Embedded Processors." In *Proceedings of the International Conference on Compilers, Architecture, Synthesis for Embedded Systems (CASES'04)*, pp. 292 - 302, Washington D.C., September 2004. (acceptance rate=25.2%, 31/123) . **(Best Paper Award of CASES 2004.)**
- SA20. Weidong Shi, Hsien-Hsin S. Lee, Chenghuai Lu, and Mrinmoy Ghosh, "Toward the Issues in Architectural Support for Protection of Software Execution." In *Workshop on Architectural Support for Security and Anti-Virus (WASSA) in conjunction with the 11th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS-XI)*, pp.1-9, Boston, Massachusetts, October 2004.

FPGA

- FP1. Taeweon Suh, Shih-Lien L. Lu, and Hsien-Hsin S. Lee, "An FPGA Approach to Quantifying Coherence Traffic Efficiency on Multiprocessor Systems" In *Proceedings of the 17th IEEE International Conference on Field Programmable Logic and Applications (FPL 2007)*, pp. 47-53, Amsterdam, The Netherlands, August, 2007. (acceptance rate = 21%) **(Nominated for the Best Paper Award.)**
- FP2. Taeweon Suh, Hsien-Hsin S. Lee, Shih-Lien Lu, and John Shen, "Coherence Traffic Considered Harmful - An FPGA Approach to Quantifying Coherence Traffic Efficiency on Multiprocessor Systems," In *International Symposium on Field-Programmable Gate Arrays (FPGA 2007)*, Monterey, California, February 2007. (poster paper.)
- FP3. Taeweon Suh, Hsien-Hsin S. Lee. Shih-Lien Lu, and John Shen, "Initial Observations of Hardware/Software Co-Simulation using FPGA in Architecture Research," In *Workshop on Architecture*

Research using FPGA Platforms (WARFP-2006) in conjunction with International Symposium on High-Performance Computer Architecture (HPCA-12), Austin, Texas, 2006.

- FP4. Taeweon Suh, Hsien-Hsin S. Lee, Sally A. McKee, and Martin Schulz, "Evaluating System-wide Monitoring Capsule Design Using Xilinx Virtex-II Pro FPGA," In *Workshop on Architecture Research using FPGA Platforms (WARFP-2005) in conjunction with International Symposium on High-Performance Computer Architecture (HPCA-11)*, San Francisco, CA, 2005.
- FP5. Martin Schulz, Brian S. White, Sally A. McKee, Hsien-Hsin S. Lee, and Jurgen Jeitner, "Owl: Next Generation System Monitoring." In *Proceedings of the ACM Computing Frontiers 2005*, pp.116-124, Ischia, Italy, May 2005.
- FP6. Martin Schulz, Brian White, Sally A. McKee, and Hsien-Hsin Lee, "A Vision for Next Generation System Monitoring." In *Workshop on Hardware Performance Monitor Design and Functionality in conjunction with International Symposium on High-Performance Computer Architecture (HPCA-11)*, San Francisco, CA, 2005.
- FP7. Christopher R. Clark, Ripal Nathuji, and Hsien-Hsin S. Lee, "Using an FPGA as a Prototyping Platform for Multi-core Processor Applications," In *Workshop on Architectural Research using FPGA Platforms (WARFP-2005) in conjunction with International Symposium on High-Performance Computer Architecture (HPCA-11)*, San Francisco, CA, 2005.

Performance Evaluation

- PE1. Richard M. Yoo, Hsien-Hsin S. Lee, Han Lee, and Kingsum Chow, "Hierarchical Means: Single Number Benchmarking with Workload Cluster Analysis." In *Proceedings of the 2007 IEEE International (IISWC-2007)*, pp.204-213, Boston, MA, September, 2007.
- PE2. Hsien-Hsin Lee, Youfeng Wu, and Gary Tyson, "Quantifying Instruction-Level Parallelism Limits on an EPIC Architecture." In *Proceedings of the IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS-2000)*, pp.21-27, Austin, Texas, April, 2000.
- PE3. Eric L. Boyd, Waqar Azeem, Hsien-Hsin Lee, Tien-Pao Shih, Shih-Hao Hung, and Edward S. Davidson, "A Hierarchical Approach to Modeling and Improving the Performance of Scientific Applications on the KSR1." In *Proceedings of the 1994 International Conference on Parallel Processing (ICPP-94)*, pp.188-192, St. Charles, Illinois, August, 1994.

EDITOR-IN-CHIEF ARTICLES

- EIC1. Hsien-Hsin S. Lee. "The AI-Hardware Flywheel." In *IEEE MICRO*, Vol.46, 3, May/June, 2026.
- EIC2. Hsien-Hsin S. Lee. "Reframing AI Memory." In *IEEE MICRO*, Vol.46, 2, March/April, 2026.
- EIC3. Hsien-Hsin S. Lee. "Silicon Pact: Reshoring the Supply Chain as Geopolitical Insurance." In *IEEE MICRO*, Vol.46, 1, Jan/Feb, 2026.
- EIC4. Hsien-Hsin S. Lee. "33-GW Pilgrimage to the Promised Land of Agent Workforce." In *IEEE MICRO*, Vol.45, 6, Nov/Dec, 2025.
- EIC5. Hsien-Hsin S. Lee. "United Semiconductors of America." In *IEEE MICRO*, Vol.45, 5, Sept/Oct, 2025.
- EIC6. Hsien-Hsin S. Lee. "Intelligence for Sale." In *IEEE MICRO*, Vol. 45, Issue 4, July/Aug, 2025.
- EIC7. Hsien-Hsin S. Lee. "Toward Disaggregated and Heterogenous AI Systems." In *IEEE MICRO*, Vol. 45, Issue 3, May/June, 2025.
- EIC8. Hsien-Hsin S. Lee. "Taiwan Semiconductor Manufacturing Company's \$165 Billion Bet." In *IEEE MICRO*, Vol. 45, Issue 2, March/April, 2025.
- EIC9. Hsien-Hsin S. Lee. "Rise of the Agentic AI Workforce." In *IEEE MICRO*, Vol. 45, Issue 1, Jan/Feb, 2025.
- EIC10. Hsien-Hsin S. Lee. "Strategic Pivot of Long-Standing x86 Rivals." In *IEEE MICRO*, Vol. 44, Issue 6, Nov/Dec, 2024.

- EIC11. Hsien-Hsin S. Lee. "The Path to Powering Intelligence." In *IEEE MICRO*, Vol. 44, Issue 5, May/June, 2024.
- EIC12. Hsien-Hsin S. Lee. "Top Picks Ignite Innovation." In *IEEE MICRO*, Vol. 44, Issue 4, July/Aug, 2024.
- EIC13. Hsien-Hsin S. Lee. "An Incoming World of Decoupling Siliconomy." In *IEEE MICRO*, Vol. 44, Issue 3, May/June, 2024.
- EIC14. Hsien-Hsin S. Lee. "Beyond Wires: The Future of Interconnects." In *IEEE MICRO*, Vol. 44, Issue 2, March/April, 2024.
- EIC15. Hsien-Hsin S. Lee. "Computing with COOL Chips." In *IEEE MICRO*, Vol. 44, Issue 1, Jan/Feb, 2024.

BLOG ARTICLES

- BL1. Carole-Jean Wu, David Brooks, Udit Gupta, Hsien-Hsin Sean Lee, and Kim Hazelwood. "[Deep Learning: It's Not All About Recognizing Cats and Dogs](#)", ACM SIGARCH Blog, November, 2019.
- BL2. Hsien-Hsin Sean Lee. "[The First Trillion-Transistor Chip: A New Design Space](#)", ACM SIGARCH Blog, August, 2019.